

The Arche Selector Operator

A Formal Framework for Discrete State Transitions

Bob Kowalski

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Abstract

This paper introduces the Arche Selector operator, a multivalued selector on a finite state space S . Given a current state s , the operator chooses among admissible successors $\mathcal{L}(s)$ those that maximize a scalar objective functional $F: S \rightarrow \mathbb{R}$. The operator is defined axiomatically and studied independently of any particular interpretation. The framework is shown to guarantee perpetual motion, finite-state recurrence, and invariance under natural transformations of the state space. A central result establishes that, for any fixed tie-breaking policy, the operator induces a partition of the state space into basins of attraction, each converging to a unique terminal strongly connected component. The operator is offered as a mathematical object applicable to discrete optimization, automata theory, cognitive architectures, and informational physics. Its compatibility with presentist ontologies is discussed as a possible interpretation.

Keywords: *discrete state transitions, selector operator, finite-state dynamics, greedy selection, graph recurrence, attractor basins*

1. Introduction

This paper introduces the Arche Selector operator $\mathcal{A}$. It is a multivalued selector on a finite state space S . Given a current state s , it chooses among admissible successors those that maximize a scalar functional $F(s)$.

While greedy algorithms are usually defined procedurally, specifying both the neighborhood structure and the selection rule within a single implementation, the Arche Selector isolates the selector function itself as an independent mathematical object. This separation allows the systematic study of the interplay between admissibility structure ($\mathcal{L}$) and objective function (F) independently of any particular implementation strategy.

The term “Selector” emphasizes this distinction: the operator selects among admissible successors, but admissibility and selection are defined separately. This separation is the defining characteristic of the framework.

The operator is defined purely mathematically. No interpretation is required for its definition or its properties. Possible applications include discrete optimization problems, automata theory and graph dynamics, cognitive architectures, and informational physics and presentist ontologies.

The choice of interpretation is left to the reader.

2. Axiomatic Definition

Axiom 1 (Finite State Space).

Let S be a non-empty finite set. Elements of S are called *states*.

Axiom 2 (Admissibility Correspondence).

$$\mathcal{L}: S \rightarrow 2^S \setminus \{\emptyset\}$$

be a function that assigns to each state s a non-empty set of admissible successors. The directed graph $(S, \mathcal{L})$ has no isolated vertices.

Axiom 3 (Objective Functional).

$$F: S \rightarrow \mathbb{R}$$

be a scalar function. Its specific form is not specified; it is a parameter of the model. The symbol F denotes “objective value” or “score”; no particular interpretation, such as complexity, energy or utility, is assumed.

Definition 1 (Arche Selector Operator).

The Arche Selector operator $\mathcal{A}: S \rightarrow 2^S \setminus \{\emptyset\}$ is defined as:

$$\mathcal{A}(s) = \arg \max_{s' \in \mathcal{L}(s)} F(s')$$

That is, $\mathcal{A}(s)$ is the set of admissible successors that attain the maximal value of F .

Definition 2 (Tie-Breaking Policy).

A tie-breaking policy is a function

$$\tau: 2^S \setminus \{\emptyset\} \rightarrow S$$

such that $\tau(A) \in A$ for every non-empty $A \subseteq S$. A deterministic trajectory is generated by composing $\mathcal{A}$ with a tie-breaking policy:

$$s_{n+1} = \tau(\mathcal{A}(s_n))$$

The framework is neutral with respect to the choice of τ . As Theorem 7 shows below, some downstream results depend on this choice whenever $\mathcal{A}$ is not single-valued.

3. Fundamental Theorems

Theorem 1 (Perpetual Walk).

For every $s \in S$, $\mathcal{A}(s) \neq \emptyset$. Therefore, every trajectory $(s_n)_{n \in \mathbb{N}}$ is an infinite walk on the finite directed graph $(S, \mathcal{L})$.

Proof. Immediate from Axiom 2 and Definition 1. ■

Theorem 2 (Finite Recurrence).

Every infinite walk on a finite directed graph eventually enters a *terminal strongly connected component* (a strongly connected component with no outgoing edges to other components) and remains within it thereafter.

Proof. In a finite directed graph, consider the set of vertices visited infinitely often along the walk; this set is non-empty, since the vertex set is finite. If there were an edge from this set to a vertex outside it, that outside vertex would also be visited infinitely often, since the edge itself would be traversed infinitely often. Hence the set has no outgoing edges. It is strongly connected, since for any two vertices u, v in the set, the walk reaches u and later v , giving a path from u to v , and symmetrically from v to u . Hence it is a terminal strongly connected component. ■

Theorem 3 (Greedy Local Optimality).

For every n , the successor selected by the operator satisfies

$$F(s_{n+1}) = \max_{s' \in \mathcal{L}(s_n)} F(s')$$

that is, s_{n+1} is a best admissible successor of s_n under F . This does not in general imply $F(s_{n+1}) \geq F(s_n)$, since s_n need not belong to its own admissible successor set $\mathcal{L}(s_n)$.

Proof. Immediate from Definition 1. ■

Corollary 1 (Monotonicity Under Self-Loops).

If $(s, s) \in \mathcal{L}$ for every $s \in S$, then $F(s_{n+1}) \geq F(s_n)$ for every n , with equality if and only if s_n already maximizes F over $\mathcal{L}(s_n)$.

Proof. Since $s_n \in \mathcal{L}(s_n)$ by hypothesis, $F(s_n)$ is among the values over which the maximum in Definition 1 is taken, so $F(s_{n+1}) = \max_{s' \in \mathcal{L}(s_n)} F(s') \geq F(s_n)$. ■

Without this hypothesis the bound fails: self-loops are usually excluded in applications (Property 1), so monotonic ascent should not be assumed by default. Examples 1 and 3 below both exhibit forced transitions in which F decreases.

Theorem 4 (Invariance Under Monotonic Transformations).

Let $\varphi: \mathbb{R} \rightarrow \mathbb{R}$ be a strictly increasing function. If $F' = \varphi \circ F$, then $\mathcal{A}$ is unchanged:

$$\arg \max_{s' \in \mathcal{L}(s)} F'(s') = \arg \max_{s' \in \mathcal{L}(s)} F(s')$$

Proof. Strictly increasing transformations preserve the ordering of values. ■

Theorem 5 (Invariance Under State Renaming).

Let $\pi: S \rightarrow S'$ be a bijection. Define $\mathcal{L}'(\pi(s)) = \pi(\mathcal{L}(s))$ and $F'(\pi(s)) = F(s)$. Then:

$$\pi(\mathcal{A}(s)) = \mathcal{A}'(\pi(s))$$

Proof. The operator depends only on the graph structure and the ordering of F , not on the labels of states. ■

Theorem 6 (Generalization to Countable State Spaces).

If S is countably infinite and $\mathcal{L}(s)$ is finite for every s , then: (1) perpetual walks still exist; (2) recurrence is not guaranteed, the walk may escape to infinity; (3) the operator remains well defined.

Proof. If $\mathcal{L}(s)$ is finite, the arg max is taken over a finite set, so it is always well defined regardless of the cardinality of S . However, when S is infinite, the walk need not return to any vertex; it may drift indefinitely, so recurrence is not guaranteed. ■

Theorem 7 (Attractor Basin Partition, Relative to a Fixed Tie-Break).

Fix a tie-breaking policy τ , inducing for each $s \in S$ the deterministic trajectory $s = s_0, s_1, s_2, \dots$ with $s_{n+1} = \tau(\mathcal{A}(s_n))$. Define an equivalence relation $\sim_\tau$ on S by $s \sim_\tau t$ if and only if the τ -trajectories from s and t eventually enter the same terminal strongly connected component. Then:

$\sim_\tau$ partitions S into basins of attraction $\{B_\tau(C)\}$.

Each basin $B_\tau(C)$ has a unique terminal SCC C such that every τ -trajectory starting in $B_\tau(C)$ eventually reaches C .

The partition is determined by the triple $(\mathcal{L}, F, \tau)$. When $\mathcal{A}$ is single-valued at every state, no ties occur, τ plays no role, and the partition is determined by $(\mathcal{L}, F)$ alone; when ties occur, different choices of τ may induce different partitions, since tied successors may lie in different basins.

The τ -trajectory from s to C need not be F -monotonic in general, as Examples 1 and 3 illustrate; by Corollary 1, it is F -monotonic whenever $\mathcal{L}$ contains a self-loop at every state.

Proof. Fix τ . By Theorem 1, the τ -trajectory from any s is a well-defined infinite walk, and by Theorem 2 it enters a unique terminal SCC $C(s)$ and remains there thereafter; uniqueness holds because the trajectory is a single deterministic sequence once τ is fixed. Define $B_\tau(C) = \{s \in S / \text{the } \tau\text{-trajectory from } s \text{ eventually enters } C\}$. Since every s has exactly one value $C(s)$, the sets $B_\tau(C)$ are pairwise disjoint and cover S , giving the partition. Determinacy from $(\mathcal{L}, F, \tau)$ is immediate, since the trajectory is an explicit iterated function of these three objects; the

qualification regarding ties follows because $\mathcal{A}(s)$ may contain more than one state of maximal F -value, and τ alone determines which is taken. ■

Corollary 2 (Cycle Structure).

If a deterministic tie-breaking policy is fixed, the trajectory eventually enters a directed cycle of states. If the tie-breaking policy is non-deterministic across visits, the trajectory is confined to a recurrent set but may be aperiodic.

Proof. Follows from Theorem 2 and the standard theory of finite directed graphs. ■

Corollary 3 (Basin Membership).

Fix τ . For any state s , the basin $B_\tau(s)$ is the set of all states whose τ -trajectory converges to the same terminal SCC as the τ -trajectory from s . This set can be characterized algorithmically by simulating the τ -trajectories.

Proof. Immediate from Theorem 7. ■

4. Properties and Remarks

Property 1 (Self-Loop Exclusion).

The definition does not require $(s, s) \notin \mathcal{L}(s)$, but such loops are usually excluded in applications to avoid trivial dynamics. As Corollary 1 shows, this exclusion is precisely what allows the objective F to decrease along a trajectory.

Property 2 (Deterministic Limit).

If $\mathcal{A}(s)$ is a singleton for every s , the operator becomes a deterministic function $S \rightarrow S$, and the dynamics reduces to iteration of a finite map. In this case no ties occur, and the basin partition of Theorem 7 no longer depends on the choice of τ .

Property 3 (Tie-Breaking as Part of the Framework).

The pair $(\mathcal{A}, \tau)$ defines a complete deterministic system. The framework is agnostic about the choice of τ , different choices yield different trajectories, and, when ties occur, potentially different basin partitions, all consistent with the axioms.

Property 4 (Parameterization).

The symbol $\mathcal{A}$ denotes the operator. If a parameter is needed for specific applications, it can be introduced as a subscript $\mathcal{A}_\alpha$, where α may influence $\mathcal{L}$ or F . The core theory is independent of such parameters.

5. Examples

Example 1 (Four States)

Let $S = \{A, B, C, D\}$. Let $F(A) = 1, F(B) = 5, F(C) = 3, F(D) = 2$, and

$$\mathcal{L}(A) = \{B, C\}, \mathcal{L}(B) = \{D\}, \mathcal{L}(C) = \{B\}, \mathcal{L}(D) = \{A, C\}.$$

The Arche Selector operator yields $\mathcal{A}(A) = \{B\}$ (since $F(B) = 5 > F(C) = 3$), $\mathcal{A}(B) = \{D\}$, $\mathcal{A}(C) = \{B\}$, and $\mathcal{A}(D) = \{C\}$ (since $F(C) = 3 > F(A) = 1$).

None of these sets has more than one element, so no tie-break choice is actually exercised in this example; the trajectory $A \rightarrow B \rightarrow D \rightarrow C \rightarrow B \rightarrow D \rightarrow C \rightarrow \dots$ is the same for every τ , entering the cycle $B \rightarrow D \rightarrow C \rightarrow B$. The single basin is $B_\tau(C_B) = \{A, B, C, D\}$, the entire state space.

This example also illustrates the limit of Theorem 3: the transition $B \rightarrow D$ takes F from 5 down to 2, since D is the only admissible successor of B and $B \notin \mathcal{L}(B)$. The objective functional is therefore not monotonic along this trajectory, consistent with Corollary 1: no self-loop is present, so the bound does not apply.

Example 2 (Hypercube, Hamming Weight)

Let $S = \{0, 1\}^N$, $\mathcal{L}(s) = \{s \oplus e_k / k = 1, \dots, N\}$, and $F(s) = \sum s_i$ summed over $i = 1$ to N .

The trajectory ascends monotonically to the maximum $11 \dots 1$, then oscillates between $11 \dots 1$ and states with one zero bit. The basin of attraction is the entire hypercube, converging to the two-state cycle at the maximum, for every choice of τ , since F strictly increases along any single-bit flip from 0 to 1, leaving no ties among improving moves except at the boundary between the two top states.

Unlike Example 1, monotonic ascent holds here even without self-loops, because every state other than $11 \dots 1$ has at least one neighbour with strictly greater Hamming weight, so the maximum over $\mathcal{L}(s)$ is never smaller than $F(s)$. This shows that the hypothesis of Corollary 1 is sufficient but not necessary for monotonicity.

Example 3 (Multiple Basins)

Let $S = \{A, B, C, D, E, F\}$. Let $F(A) = 1, F(B) = 5, F(C) = 4, F(D) = 3, F(E) = 2, F(F) = 1$, and

$$\mathcal{L}(A) = \{B\}, \mathcal{L}(B) = \{C\}, \mathcal{L}(C) = \{B\} \text{ (cycle } B \leftrightarrow C\text{), and } \mathcal{L}(D) = \{E\}, \mathcal{L}(E) = \{F\}, \mathcal{L}(F) = \{E\} \text{ (cycle } E \leftrightarrow F\text{)}.$$

As in Example 1, every admissible successor set is a singleton, so τ plays no role. The basins are $B_\tau(C_1) = \{A, B, C\}$, converging to the cycle $\{B, C\}$, and $B_\tau(C_2) = \{D, E, F\}$, converging to the cycle $\{E, F\}$, illustrating Theorem 7 with two distinct terminal components.

This example gives a second, independent counterexample to unconditional monotonic ascent: $B \rightarrow C$ takes F from 5 down to 4, and $E \rightarrow F$ takes F from 2 down to 1, both forced moves with no self-loop available.

6. Possible Interpretations

The Arche Selector operator is compatible with several interpretations.

- **Presentist Ontology.** If S is interpreted as the set of possible physical states and only the current state is assumed to exist, the operator provides a deterministic rule for state succession without memory. The attractor basins correspond to regions of state space from which the dynamics converges to a particular cyclic behaviour.
- **Discrete Optimization.** The operator is a greedy local search method that always moves to the best neighbour according to F . The basin partition describes the regions of the search space that converge to each local optimum.
- **Cognitive Architectures.** The operator models a reactive system that selects actions based solely on the current state and a fixed objective. Basins correspond to behavioural modes.
- **Informational Physics.** The operator models a computational substrate that updates a finite informational state. Basins represent regions of the informational state space with stable long-term dynamics.

No interpretation is required for the mathematical validity of the framework.

7. Open Problems

1. Characterization of recurrent sets. Given $\mathcal{L}$ and F , can the terminal SCCs be characterized a priori? What conditions on $\mathcal{L}$ and F guarantee a unique basin?
2. Complexity of the operator. What is the computational cost of evaluating $\mathcal{A}(s)$ for arbitrary $\mathcal{L}$ and F ?
3. Continuous limits. Can the discrete operator be extended to continuous state spaces? Under what conditions does the basin partition have a continuous analogue?
4. Stochastic extensions. What happens if τ is chosen randomly? Does the basin partition remain well-defined in expectation?
5. Coupled systems. How do multiple interconnected Arche Selector operators behave? Can basin structures interact?
6. Reachability. Given two states s, t , can we determine whether they belong to the same basin without simulating the trajectory?

7. Robustness under tie-breaking. When $\mathcal{A}$ is not single-valued, how does the basin partition vary across different choices of τ , and can classes of τ be characterized that yield the same partition?

8. Conclusion

The Arche Selector operator $\mathcal{A}$ is a formally defined, axiomatically grounded multivalued selector on finite state spaces. It guarantees perpetual walks and eventual recurrence into a terminal strongly connected component, and it selects at each step a locally optimal admissible successor with respect to the objective functional F , with true monotonic ascent holding under the additional hypothesis of self-loops. The central result, the Attractor Basin Partition Theorem, establishes that, for any fixed tie-breaking policy, the operator induces a partition of the state space into basins, each converging to a unique terminal component; this partition is determined by $(\mathcal{L}, F)$ alone when no ties occur, and depends additionally on the tie-breaking policy otherwise. The framework is invariant under state renaming and monotonic transformations of the objective. It generalizes to countable state spaces when the admissibility sets are finite. The operator is offered as a mathematical object independent of interpretation, applicable to optimization, automata, cognitive modeling, and informational physics. Its compatibility with presentist ontologies is noted as one possible application.

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